Project Design Phase
Solution Architecture

Date	16 February 2026
Team ID	LTVIP2026TMIDS54324
Project Name	Smart Sorting: Identifying rotten fruits and vegetables using transfer learning
Maximum Marks	4 Marks

Solution Architecture: The Smart Sort Application mainly consists of: Core Components

1. **Frontend (Presentation Layer)** ○ Built with HTML, CSS, JavaScript ○ Allows users to upload or drag-and-drop images ○ Displays prediction results (label + confidence)
2. **Backend (Application Logic)** ○ Developed using Flask (Python) ○ Handles image reception and routing ○ Connects to the trained deep learning model for inference
3. **Model Layer (AI Engine)** ○ Uses **MobileNetV2** (transfer learning from ImageNet) ○ Preprocessing: resize (224×224), normalize, preprocess input ○ Output: 28-class softmax prediction with high accuracy
4. **Data Layer** ○ Input dataset from Kaggle (28 classes: fresh & rotten produce) ○ Images are augmented and standardized before training ○ .h5 model file packaged with the application
5. **Deployment Layer** ○ App bundled as .exe using PyInstaller for offline use ○ Runs on standalone systems without needing Python ○ Folder structure designed for portable execution

🔁 End-to-End Flow

User Upload → Flask API → Preprocess → Model Predict → Result Return → UI Render

Solution Architecture Diagram:

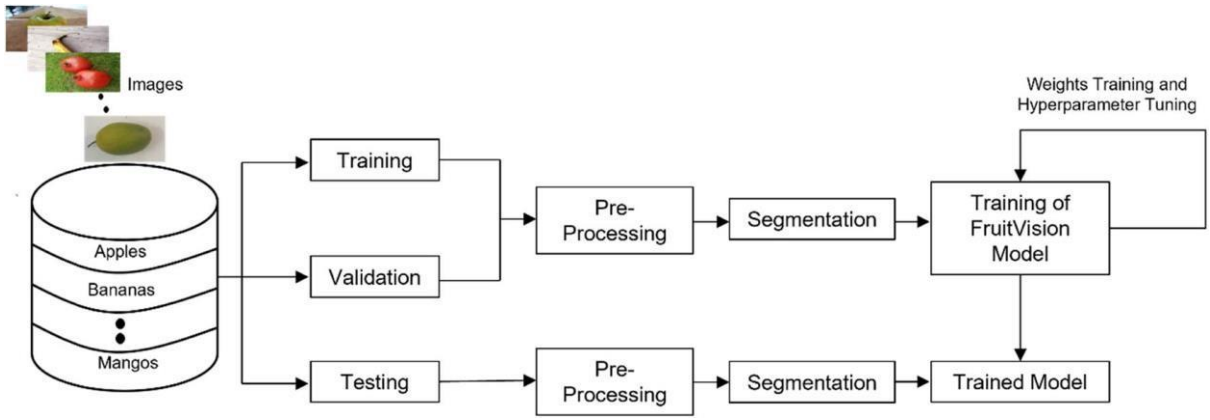

Figure 1: Architecture and data flow of the Model

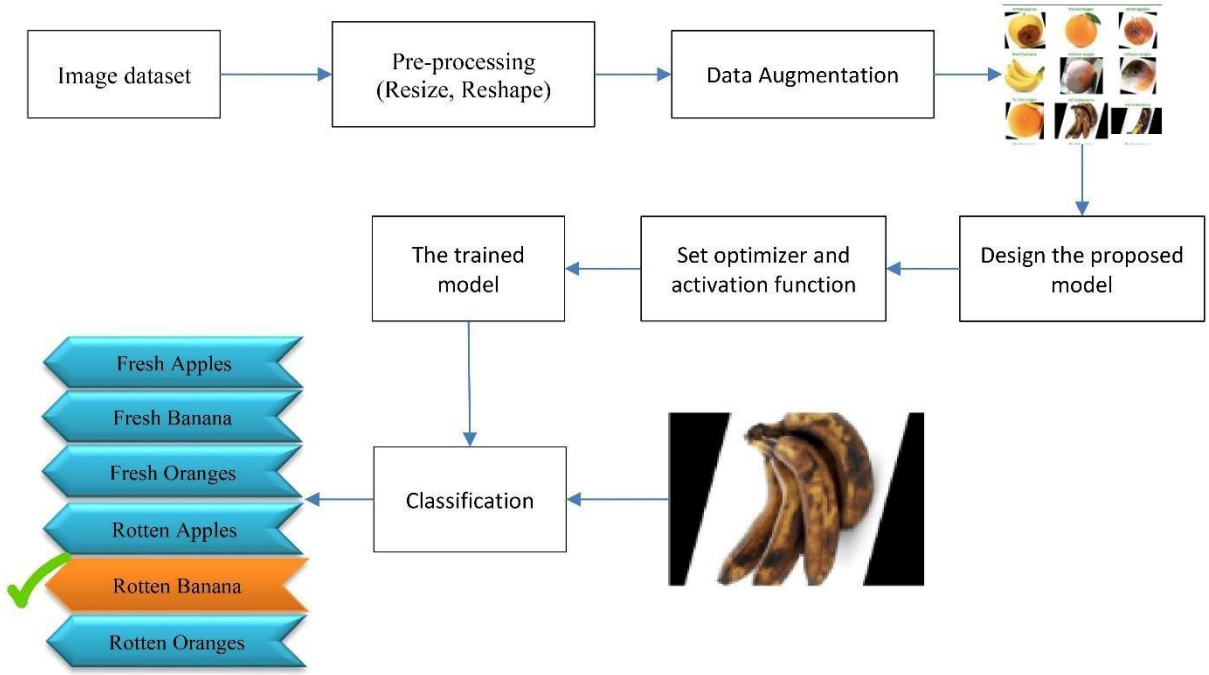

Figure 2: Architecture and data flow of the Smart Sort Application